

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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Register Number	192525076

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **R.Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *R. Pranathi*

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

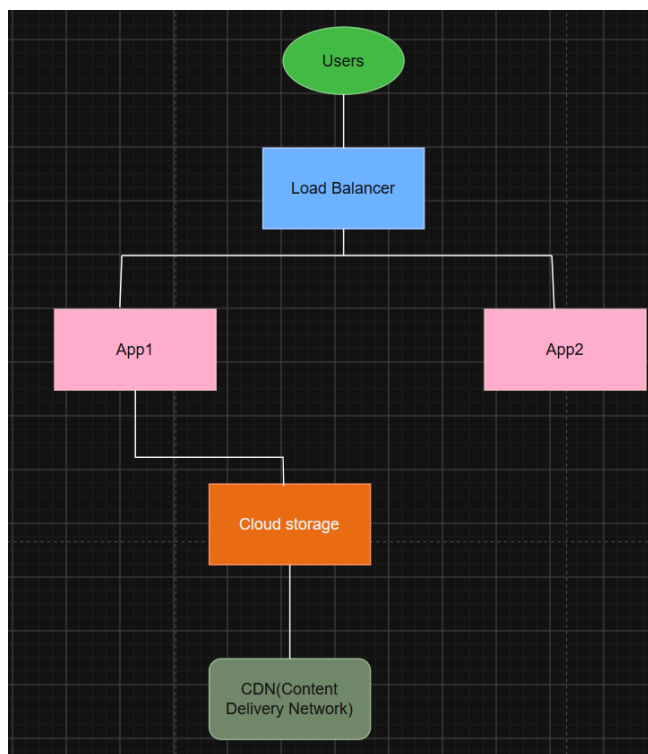
1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

ANS:

When a media company moves its video streaming platform to the cloud, it should design a system that can support millions of users at the same time without buffering. The cloud architecture should include a load balancer, application servers, cloud storage, a Content Delivery Network (CDN), and auto-scaling services. The load balancer distributes user requests to different servers so that no single server becomes overloaded. Videos are stored in cloud storage, while the CDN keeps copies of popular videos closer to users. This helps reduce buffering and improves streaming speed.

Auto-scaling is another important feature because it automatically increases the number of servers when user traffic is high and reduces them when traffic becomes low. This ensures that users always receive a smooth streaming experience while avoiding unnecessary resource usage. Cloud monitoring services continuously check CPU usage, memory, and server health so that any problem can be detected early.

To reduce operational costs, the company can use the pay-as-you-go model, where it only pays for the resources it actually uses. Frequently watched videos can be cached in the CDN, reducing the costs. Old or rarely watched videos can be moved to cheaper cloud storage. Reserved instances and spot instances can also be used to lower infrastructure expenses. Overall, this cloud architecture provides high scalability, better performance, low latency, cost savings, and highly available, making it suitable for large-scale video streaming platforms.

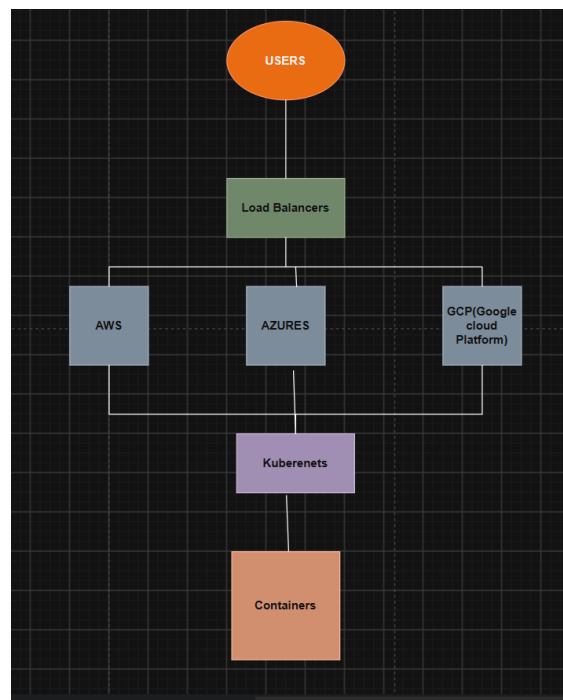


2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

ANS:

A fintech startup that uses multiple cloud providers should deploy its applications using containers because containers make applications portable and easy to manage. Each container includes the application along with all the required software and libraries, so it works the same way on every cloud platform. To manage these containers efficiently, Kubernetes can be used as a container orchestration tool. It automatically deploys, monitors, scales, and restarts containers whenever required, reducing manual work and improving application reliability.

A multi-cloud strategy helps the company avoid depending on a single cloud provider. If one cloud provider experiences an outage, another provider continues running the application, ensuring uninterrupted financial services. Service discovery allows different application components to communicate automatically without manually configuring IP addresses. Auto-scaling increases the number of containers during high transaction periods and decreases them when traffic is low, helping improve performance while reducing costs. Although a multi-cloud model provides high availability, flexibility, and disaster recovery, it also requires proper monitoring, security management, and synchronization between cloud providers. Overall, this solution provides better reliability, scalability, and business continuity for the fintech startup.

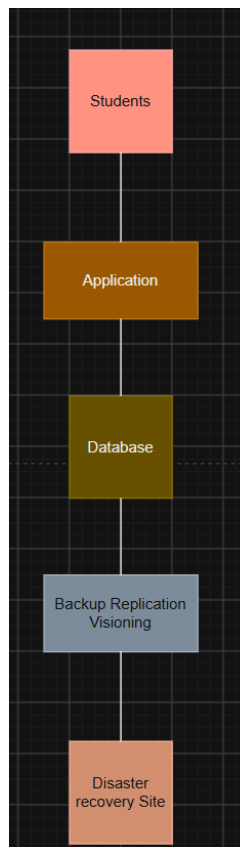


3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

ANS:

An e-learning platform should have a proper backup and disaster recovery plan to protect important data such as student records, course materials, assignments, and examination results. Cloud-native backup services automatically create copies of data at regular intervals, ensuring that important information can be recovered if files are accidentally deleted or damaged. Data replication stores copies of data in another cloud region so that services remain available even if one region experiences a failure. Versioning keeps multiple copies of files, allowing users to restore previous versions whenever necessary.

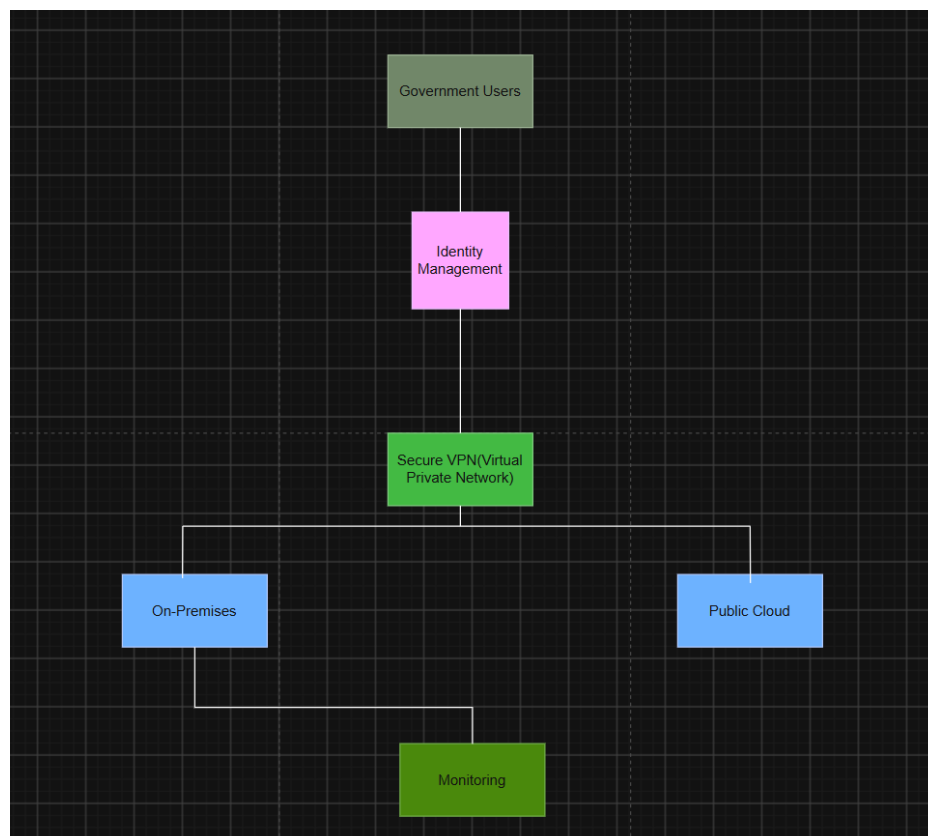
Recovery Time Objective (RTO) is the maximum time required to restore the application after a failure, while Recovery Point Objective (RPO) represents the maximum amount of data that can be lost. These objectives can be achieved through regular backups, continuous replication, and automatic failover. To maintain business continuity, the organization should monitor system performance, encrypt backup data, perform regular disaster recovery testing, and keep backup copies in different cloud locations. This strategy reduces downtime, protects valuable educational data, and ensures that online learning continues without major interruptions.



4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

A government agency can use a hybrid cloud model to securely manage both sensitive and non-sensitive information. Sensitive data such as citizen records and confidential government documents should remain in the on-premises data center, while non-sensitive applications such as public websites and online services can be hosted in the public cloud. Secure communication between these two environments can be established using VPN connections and encrypted communication protocols. This allows data to move safely without exposing it to unauthorized users.

Identity management is important because it ensures that only authorized employees can access government systems. Authentication methods such as passwords and multi-factor authentication improve security, while role-based access control gives employees access only to the information required for their work. Encryption protects data during storage and transmission, preventing unauthorized access even if the data is intercepted. Although hybrid cloud provides flexibility, scalability, and improved security, managing both cloud and on-premises environments can be complex and requires continuous monitoring and proper compliance with government regulations.



5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

ANS:

A SaaS provider should deploy its application across multiple cloud regions to ensure that services remain available even if one region fails. Users connect through a global load balancer, which distributes traffic to the nearest healthy region. If one region becomes unavailable due to hardware failure or network issues, traffic is automatically redirected to another region, allowing users to continue accessing the application without interruption. This improves availability and ensures business continuity.

Data replication keeps copies of application data in different cloud regions. Synchronous replication ensures that all copies remain identical, while asynchronous replication provides better performance with a small delay in data synchronization. Automatic failover detects failures and switches user traffic to a healthy region without manual intervention. Monitoring services continuously check server performance, CPU usage, memory utilization, network traffic, and application health. Alerting systems immediately notify administrators if any problem occurs so that it can be resolved quickly. This approach provides high availability, fault tolerance, reliable disaster recovery, and a better user experience for SaaS applications.

